

Development of plant phenotyping system using Pan Tilt Zoom camera and verification of its validity

Dong Thanh Pham^a, Nayeem AI Amin^a, Daisuke Yasutake^b, Yasumaru Hirai^b, Takenori Ozaki^c, Masaharu Koga^c, Kota Hidaka^d, Masaharu Kitano^e, Hien Bich Vo^f, Takashi Okayasu^{b,*}

^a GS of Bioresource and Bioenvironmental Sciences, Kyushu University, Japan

^b Faculty of Agriculture, Kyushu University, Japan

^c welzo Co. Ltd., Japan

^d NARO Kyushu Okinawa Agricultural Research Center, Japan

^e IoP Collaborative Creation Center, Kochi University, Japan

^f Electrical and Computer Engineering Department, Vietnamese-German University, Viet Nam

ARTICLE INFO

Keywords:

Plant phenotyping
Robotics
Deep learning
Pan Tilt Zoom camera
Precision agriculture

ABSTRACT

Quantitative analysis for plant growth attributes has gained prominence in plant science and agriculture. Despite the availability of automated phenotyping systems as a solution to labor-intensive manual measurement techniques, these systems often require specialized knowledge and face challenges in scaling for high-throughput applications. This research introduces a scalable high-throughput plant phenotyping technique utilizing a Pan Tilt Zoom (PTZ) camera. The primary objective is to assess the application of a PTZ camera in a plant phenotyping system. By integrating open-source software and hardware technologies, the method captures images of cucumber plants in a controlled greenhouse environment. The operational procedure of the robot consists of a series of steps. It begins with the robot's initial movement to capture infrared images, followed by an analysis to detect Aruco markers serving as location identifiers for capturing plant images. Subsequently, the PTZ camera is adjusted to capture specific plant traits from predefined viewpoints. The captured images with location IDs, preset viewpoints, and timestamps are then sent to a remote server. Validation of the system's dependability includes manual measurements on fundamental operations and the evaluation of the effectiveness of zoomed images captured by the PTZ camera, tested through plant feature detection. Experimental results demonstrate promising outcomes, achieving a mean average precision (mAP) of 94%, 97.6%, 98.4%, 90.1%, and 97.6% for apical buds, male flowers, female flowers, tiny cucumbers, and mature cucumbers respectively when using the trained YOLOv8s on an augmented dataset tested on highly zoomed images validation set, outperforming less zoomed or less detailed validation sets. These findings underscore the efficacy of this innovative approach in capturing real-time plant images. Leveraging the PTZ camera's zoom, pan, and tilt capabilities enables comprehensive visualization of plant traits and adaptability to evolving growth patterns, thereby improving the results of plant feature detection. The amassed imagery serves a dual purpose by acting as training data for AI models, highlighting their potential to facilitate future research endeavors demanding extensive and scalable plant information.

1. Introduction

In light of the global challenge of sustaining an ever-growing population, a report by FAO in 2018 highlights a pressing concern. To feed a projected world population of nearly 10 billion by 2050, a substantial increase of over 54% in agricultural production would be required compared to the levels of 2012 (FAO, 2018). This ongoing worldwide food crisis highlights the need for stable and high agricultural production. To tackle this challenge, quantitative analysis of phenotype development

has recently become a potential solution. High-throughput plant phenotyping (HTPP) methods for detecting plant changes based on aerial photographs could be used for large-scale surveys of agricultural lands inoculated with beneficial microorganisms (Shi et al., 2018). Besides, climate change imposes sudden fluctuations in environmental conditions (Cramer et al., 2011; Will et al., 2013) which threaten agricultural production, affecting crop yield, biomass, biochemical composition, and visual quality of plants, thus impacting the marketability of products (Rouphael et al., 2018). Phenotyping with anatomical traits of

* Corresponding author.

E-mail address: okayasu@bpes.kyushu-u.ac.jp (T. Okayasu).

<https://doi.org/10.1016/j.compag.2024.109579>

Received 23 August 2023; Received in revised form 2 February 2024; Accepted 20 October 2024

Available online 28 November 2024

0168-1699/© 2024 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

leaves can unveil crop acclimation mechanisms and predict possible physiological behaviors after sudden environmental fluctuations due to climate changes (Amitrano et al., 2022).

The integration of electrical and computer engineering and science with agronomy and life sciences has developed plant phenotyping from a traditional low-throughput, time-consuming, labor-intensive, and subjective task into one of the most advanced areas of research in crop science (Singh et al., 2021; Van Eeuwijk et al., 2019; Zhao et al., 2019).

Recent technological advances have allowed the development of HTPP systems for automatically acquiring plant traits at higher spatiotemporal densities (Yang et al., 2020). The edge of advanced systems includes different capturing stops for non-invasive measurements and computer-controlled conveyors for moving plant pots to different stations (Casto et al., 2021). While these systems can provide a rich collection of data on various phenotypes, they not only require costly hardware and software but also demand specialists for operation. To address the food crisis, the cost, and the throughput of agricultural systems, the HTPP system should also be improved (Ray et al., 2013; Prasanna et al., 2013). This drive for improvement intersects with the significance of temporal measurements in understanding phenotypic changes during plant development (Berger et al., 2012; Ge et al., 2016). Besides, compared to measurements by human observers, automated measurements can be more precise and provide spatially dense data for analysis. Therefore, the cost of HTPP systems for collecting phenotype information is often driven by the automation of movement and the measurements' temporal density, which requires specialized hardware and software. The movement of plants toward sensors using conveyors decreases the number of sensors that need to be integrated into the system (Tanabata et al., 2022). However, this method can only be applied to small-size plants. Furthermore, incorporating location data becomes pivotal in controlled environments like greenhouses, where environmental variations across different locations can significantly impact plant growth and development (Muramatsu and Kubota, 2021). By collecting location data and using it in data analysis, researchers and growers can better understand the spatial variability of these environmental factors and their impact on plant growth and development. With the use of location data, HTPP technologies such as imaging systems, drones, and sensors can be deployed in a targeted way to optimize data collection, analysis, and interpretation. This approach can support researchers and growers in identifying the best installing locations for sensors and the target monitoring areas and monitoring and developing spatially-resolved models to predict crop performance and optimize resource allocation.

Precise algorithms are crucial for efficiently conducting image analyses, enabling the extraction of valuable plant phenotyping data (Mochida et al., 2019; Su et al., 2023). Thus, enhancing the temporal density of measurements and scalability in High-Throughput Plant Phenotyping (HTPP) systems can be achieved by incorporating more sensible cameras, all without compromising data quality.

However, cameras for monitoring plant phenotypes encounter various challenges in ensuring efficient operation. The rapid growth of plant traits inside greenhouses, both at the plant and canopy levels over an extended period (days to weeks), presents a difficulty in capturing phenotypes that fall outside the camera's field of view (Hilty et al., 2021; Johnson et al., 2009). Additionally, USB cameras may struggle to collect high-quality data for small subjects of interest, a requirement in various studies focusing on leaf disease classification (Harakannanavar et al., 2022; Peng and Wang, 2022; Kumar et al., 2022) and flower detection (Mann et al., 2022).

While Hyperspectral Imaging (HSI) has emerged as a viable method for measuring traits across a broad spectrum simultaneously in 3D, enabling the definition of abiotic, biotic, and quality traits for crop management optimization (Mertens et al., 2021; Bodner et al., 2018), it comes with drawbacks. HSI demands specialized hardware, more time for data collection, increased storage space, and costly data analysis

software, which collectively limit the temporal density of measurements. In contrast, RGB images, processed through precise algorithms, may be more practical for capturing phenotypes due to the impact of leaf overlap on near-infrared reflectance (Kefauver et al., 2015). Consequently, incorporating a more significant number of sensible RGB cameras in HTPP systems can improve the temporal density of measurements and scalability without compromising data quality.

To expand the scope of monitoring, a Pan Tilt Zoom (PTZ) camera has been applied in some research. Fukada et al. (2023) presented a pan-tilt camera mount with three axes, utilizing servomotors to achieve diverse imaging capabilities with a single camera device. The servomotors are linked to a Raspberry Pi through a PWM servo driver equipped with a PCA9685 controller. Despite its cost-effectiveness, this system requires multiple cameras installed to monitor the plant canopy and lacks zoom functionality, limiting the effectiveness in capturing detailed images of plant canopies. Utilizing a PTZ camera system featuring 360° rotation and pitching, a helix four-bar mechanism for vertical movement, and a sliding block guide for horizontal motion, tomato plants were effectively monitored (Han et al., 2014). Nevertheless, the system demands extensive hardware installation and human intervention in remote management. The PTZ camera system (Kim et al., 2020) demonstrates the capability to monitor a broad area and capture periodic images, effectively detecting downy mildew in onions through a deep learning-based approach. However, a limitation of the system is its lack of movement features, restricting its ability to monitor plants around its fixed position.

Along with the development of plant monitoring systems, there has been a growing utilization of object detection algorithms based on deep learning in agriculture (Tian et al., 2020). These algorithms can be broadly classified into two categories. The first is the R-CNN (Region-based Convolutional Neural Network) family of algorithms, which are geared for regional proposals and comprise representative networks such as R-CNN, Fast R-CNN, Faster R-CNN, Mask R-CNN, etc. Another category is one-stage algorithms and their representative networks, such as the YOLO (you only look once) series. Compared to R-CNN models, the YOLO model, first introduced by Redmon et al. (2016), offers a distinct advantage by unifying target classification and localization within a regression network architecture. With advancements made in the second to eighth-generation iterations, the YOLO series has not only improved detection accuracy and speed but also provided lightweight deployment options (Redmon and Farhadi, 2017, 2018; Bochkovskiy et al., 2020; Jocher et al., 2020; Wang et al., 2022).

Comprising three core elements – input processing, backbone, and head segments – the YOLOv7 model (Wang et al., 2022) integrates advanced structural components like ELAN, MP, and SPPCSPC within the foundational CBS structure. ELAN, or Efficient Layer Aggregation Network, optimizes local information aggregation, enhancing the model's object detection in complex scenes. MP ensures robustness by compacting feature maps and emphasizing critical elements. SPPCSPC, a fusion of CSPNet with SPP block, enriches information extraction across spatial and channel dimensions, which is crucial for diverse object scales and complex scenarios. In contrast to YOLOv5 (Xiong et al., 2022), YOLOv7 introduces trainable “bag-of-freebies” algorithms, enabling real-time object detection with high accuracy and efficiency in inference costs. Incorporating insights from YOLOv5 and YOLOX (Li et al., 2022), YOLOv7 introduces the innovative Coarse-to-fine guided label allocation strategy, enhancing the model's object detection capabilities through strategic sample assignment to distinct output layers.

YOLOv8, the current state-of-the-art model, represents the latest iteration of the YOLO object detection and image segmentation model, an advancement over YOLOv5 developed by Ultralytics, as detailed by Terven and Cordova-Esparza in 2023. YOLOv8 enhances its predecessor in five key aspects: structural design, loss computation, training data augmentation, training strategy, and model inference process, making it versatile across various visual domains. In precise plant leaf image

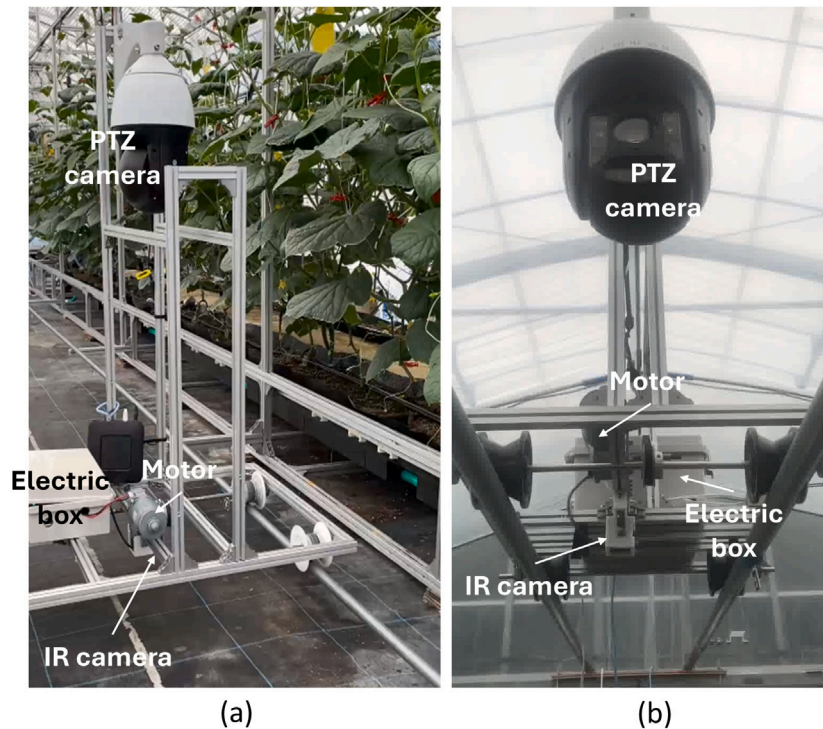


Fig. 1. General system design: (a) Ground-based robot and (b) Overhead robot.

segmentation, Yang et al. (2023) presents an inventive approach amalgamating YOLOv8 and an enhanced DeepLabv3+, achieving a noteworthy 90.8% mean intersection over union (mIoU). Similarly, Xiao et al. (2023) attains a remarkable 99.5% accuracy in classifying fruits as ripe or overripe through the utilization of YOLOv8 with CSP and C2f modules, elevating the capabilities of deep learning-based visual object detection for fruit characterization.

This paper presents the development of a scalable HTPP system capable of collecting plant feature data, and a plant feature detection model is trained using the collected data. Despite limited resources, this system accurately and automatically measures valuable plant traits non-destructively. The system consists of two digital cameras: an IR camera tracks the system's location, while a PTZ camera captures plant images from various viewpoints. A Raspberry Pi computer controls these cameras. The system's performance was tested in a cucumber greenhouse experiment, and the collected images were used to train plant feature detection.

2. Materials and methods

2.1. General system design

Fig. 1 illustrates the configuration of plant phenotyping robots equipped with PTZ cameras. Ground-based robots offer advantages such as efficient use of resources and affordability. On the other hand, overhead robots provide greater coverage and space-saving benefits. The selection between the two options depends on the specific research objectives and the availability of resources. Fig. 2 shows the electronic components for the developed plant phenotyping system. The main components were a Raspberry Pi computer (<https://www.raspberrypi.org>; Raspberry Pi Foundation, Cambridge, United Kingdom), a motor driver (Cytron), a DC motor (Enstroert, 24 V), four wheels, two reed switches, one security switch (Limit switch), an Infrared camera (IR camera) and a PTZ camera (Sunba 601). The robot frame was assembled by aluminum frames with 2 cm square cross-section. The power supply is located nearby to power the entire system.

2.1.1. Navigation system

To aid the robot's navigation, two reed switches are installed on the robot, with corresponding magnetic parts placed at the start and end of the railroad as illustrated in Fig. 4a. When the switch crosses the magnetic part, it triggers the switch and allows the vehicle to identify its direction and limit points of the road. A security switch has been implemented to prevent any issues with the reed switches on the robot, cutting off motor power if activated. The limit bars at both ends of the road activate the security switch when the vehicle touches them.

2.1.2. Application of ONVIF protocol in controlling PTZ camera

Established in 2008 by AXIS, BOSCH, and SONY, the Open Network Video Interface Forum (ONVIF) is a non-profit collaboration. It strives to standardize protocols for intelligent security network devices, boosting product compatibility. Over time, ONVIF has progressed from its V1.0 specification to the latest version, V23.06.

ONVIF is a pivotal function library in managing PTZ (pan, tilt, zoom) cameras, facilitating control over these functions through its standardized API, the ONVIF PTZ Service. The operational mechanics of ONVIF's control over these camera attributes can be delineated as follows:

In the realm of PTZ manipulation, ONVIF delineates precise commands that orchestrate the dynamic orientation of PTZ cameras. This encompasses commanding the camera to rotate horizontally (pan), adjust vertically (tilt), and modify its lens magnification (zoom). All the camera commands are conveyed via the ONVIF protocol through network communication, ensuring a uniform interpretation and response across PTZ cameras provided by different camera vendors.

Furthermore, ONVIF seamlessly integrates Preset functionality, allowing PTZ cameras to navigate to predefined positions promptly. This process involves two distinct phases: first, issuing a command to the camera to store its current configuration and position as a preset; second, instructing the camera to return to a previously saved preset viewpoint. This facet proves instrumental in automating surveillance tasks, such as effortlessly focusing on pre-designated zones or swiftly zooming in on specific points of interest.

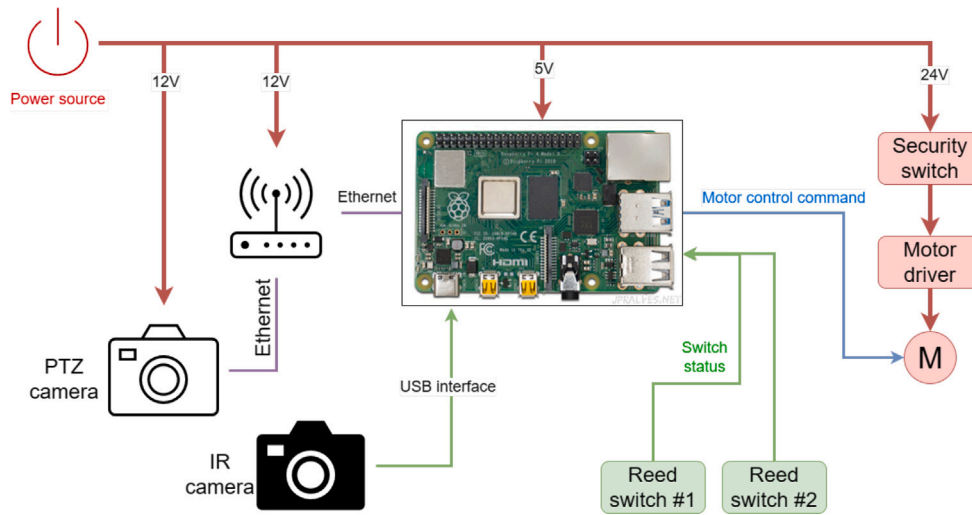


Fig. 2. Electronic components for plant phenotyping system.

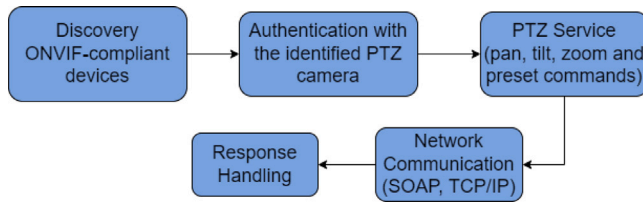


Fig. 3. Process to manage PTZ cameras through ONVIF.

To effectively manage PTZ cameras through ONVIF, the process involves several key stages as shown in Fig. 3 with control commands supported by www.onvif.org and code example of ONVIF implementation in Raspberry Pi is provided in https://github.com/dongpham46/onvif_raspberry_pi. Commencing with Discovery, the ONVIF Device Discovery protocol is utilized to locate ONVIF-compliant devices on the network. Subsequently, Authentication comes into play, necessitating the use of provided credentials to establish a connection with the identified PTZ camera. Interaction with the camera is achieved through the ONVIF PTZ Service API, which facilitates the transmission of PTZ control instructions such as panning, tilting, zooming, and preset commands like storing and recalling specific configurations. These commands are conveyed through standard network communication protocols such as SOAP (Simple Object Access Protocol) over TCP/IP (Transmission Control Protocol/Internet Protocol). Finally, Response Handling entails interpreting the camera's feedback, discerning the success of the executed actions, or identifying potential errors.

The used PTZ camera (Sunba camera), with specifications including a 5-megapixel resolution, 20x zoom, and night vision capabilities, is strategically positioned at the top of the system. The PTZ camera and the Raspberry Pi controller are connected to the same network router using Ethernet cables. The Raspberry Pi is equipped with the ONVIF protocol, allowing it to control the camera's orientation with various angles and zoom levels, which can be preset through the Raspberry Pi or the ONVIF camera software.

2.1.3. Operation procedure

Fig. 4 illustrates the robot's orientation. The "Forward" direction refers to the movement from the start to the end of the railroad, while the "Backward" direction is the opposite. The "Home" is situated between the road's beginning and end and is identified using a designated AR marker ID. Users can freely reposition this marker as required to accommodate their research requirements.

Fig. 4b and 4c delineate the operational sequence of the system. Commencing with the IR camera in motion, the robot initiates its journey. The robot stops and begins to trigger the PTZ camera's movement through ONVIF commands when the IR camera recognizes the Aruco marker. Subsequently, the process of Image acquisition commences. Besides, due to the PTZ camera's requirement for additional time to focus after robot movement, a delayed frame capture approach is adopted and tailored to this scenario. Each obtained image is archived and transmitted to the remote server, accompanied by pertinent metadata such as location, timestamp, and preset viewpoint. These iterative steps persist until reed switch #2 is activated, signifying the end of the railroad. At this juncture, the robot's trajectory is inverted, and it retraces its path to the start of the railroad. Activation of reed switch #1 prompts a return to forward motion. The image-capturing process is reinitiated until the robot reaches the designated "Home" position, where it halts, anticipating the next scheduled task.

The operation plan can be pre-scheduled based on the research requirements and remotely managed. All the commands for connecting to each camera, capturing images by the cameras, controlling the motor, recognizing all the switches' status, and processing images were executed by python scripts in the Raspberry Pi. To automatically schedule monitoring tasks for the system, connecting to a local network the system was provided actual time for execute each script.

2.1.4. Alert function

The robot is outfitted with a pair of distinct alert mechanisms. The initial mechanism swiftly notifies any motor operation failures that might transpire during routine activities. This is achieved by assessing the most recent AR marker ID. This assessment helps determine whether the robot has reached the designated Home point. The second alert mechanism sends notifications when the PTZ camera encounters difficulties capturing images of plant features. This is ascertained by monitoring the quantity of captured images at each collection location, which allows the system to detect any potential failures in PTZ camera movement and image capture. The system is meticulously designed in both scenarios to dispatch timely notifications to a preconfigured email address.

2.2. Data acquisition

A visual representation of the image acquisition procedure throughout the study is illustrated in Figs. 5 and 6. The ground-based robot utilizes a PTZ camera positioned 0.7 meters horizontally from the soil, maintaining a height of 1.3 meters above the ground. It captures images

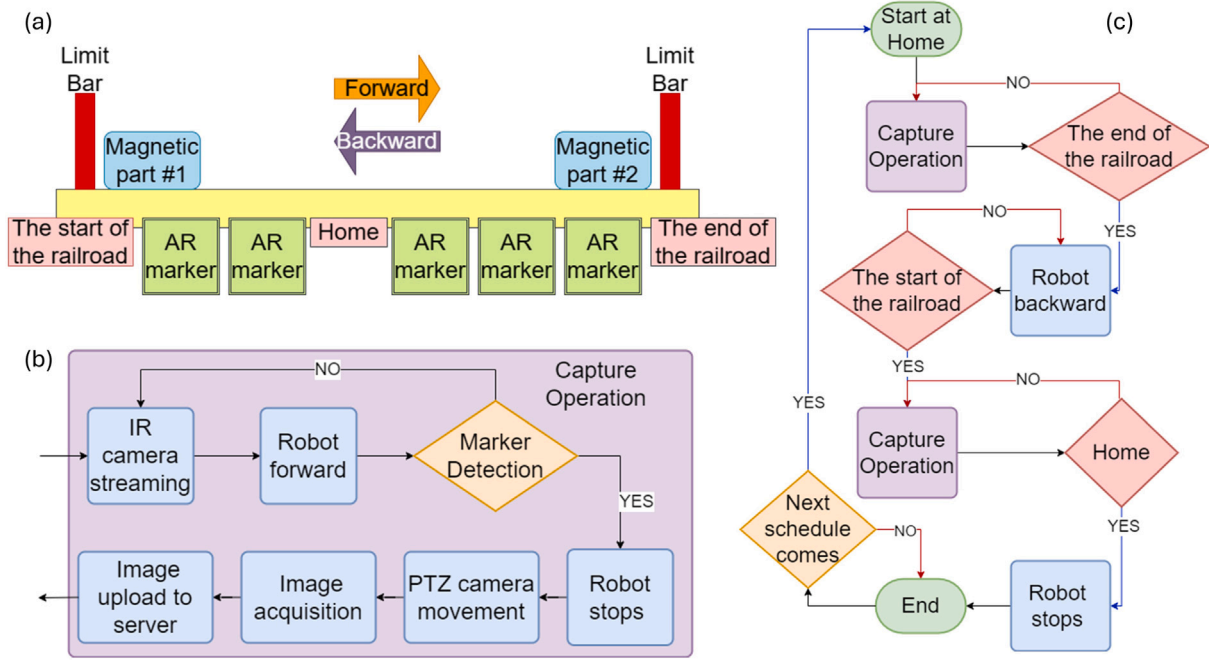


Fig. 4. The system operation procedure. (a) The movement direction of the robot. (b) The Capture Operation. (c) The General Operation Procedure.

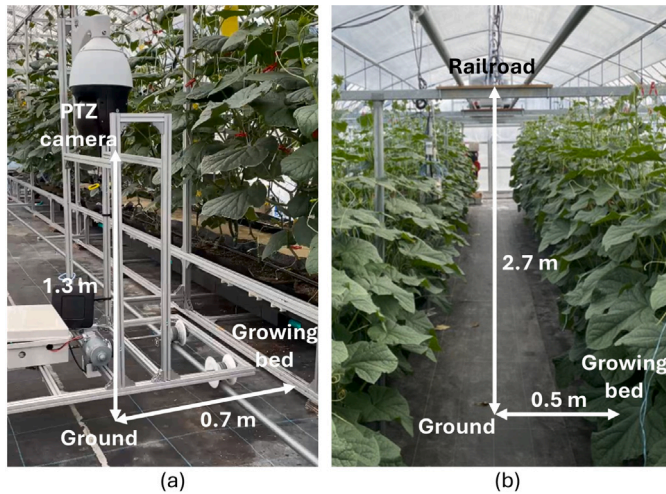


Fig. 5. The camera locations: (a) Ground-based robot, (b) Overhead robot.

as it moves forward to the end limit of the railroad, then reverses back to the starting point and returns to the home position. The captured plants are on one side of the railroad. On the other hand, the overhead robot has a PTZ camera situated horizontally 0.5 meters from the soil and at a height of 2.7 meters above the ground. It captures images while moving forward to the end, backward to the start of the railroad, and forward to the home position. The plants are captured on both sides of the railroad and only during forward movement. The RGB images collected have dimensions of 2592×1944 pixels and are saved in jpg format.

2.2.1. Location data acquisition

The system utilized an IR camera positioned at the bottom to detect Aruco markers with a unique identifier (ID) connected to the measurement locations. The Aruco markers comprise a seven-by-seven binary grid generated by an open-source software dictionary and placed

beneath the railroad as illustrated in Fig. 7. The ELP 2 Megapixel WEB Camera with Night Vision IR CUT Infrared is utilized to detect the Aruco markers with the camera's infrared filter functioning continuously to ensure daily and night identification. To identify the capture location and also the location of the moving system, the system streams one side of the railroad via the IR camera, and as the system moves, AR markers appear in the streaming. They are identified by the computer vision package, OpenCV. Next, the ID information generated from the AR markers is checked to ensure that it is distinct from the previous ID and within the preset ID list range to minimize false detection. Finally, the system halted, recorded the ID information, and began the capture operation. Notably, for enhanced versatility and usability by farmers and researchers, the AR markers are equipped with magnetic components, enabling free movement along the railroad. This flexibility allows images to be captured precisely at desired locations, facilitating ease and convenience in various agricultural and research applications.

2.2.2. Dataset preparation

The chosen classes for dataset preparation include "apical bud", "male flower", "female flower", "tiny cucumber", and "mature cucumber". The apical bud, situated at the pinnacle of the main stem, is instrumental in directing vertical growth by producing auxin. A hormone suppresses the development of lateral buds along the stem. As the plant matures, it transitions to the flowering stage. Female flowers, identified by a small cucumber-like structure at their base, transform into tiny cucumbers upon successful pollination. While male cucumber flowers are smaller with slender structures, lacking a swollen base resembling a tiny cucumber and are often found in clusters along the main stem and branches of the cucumber plant. The tiny cucumbers undergo gradual growth and maturation, producing mature cucumbers ready for harvest.

The labeling annotation tool is utilized to outline bounding boxes, which encapsulate specific objects of interest. Each bounding box is associated with an annotation file that records essential details, including the coordinates of the bounding box's center and its width and height dimensions. In Fig. 8, examples of labeled classes are depicted, showcasing bounding boxes that encompass the entirety of the object content and closely adhere to the object boundary.

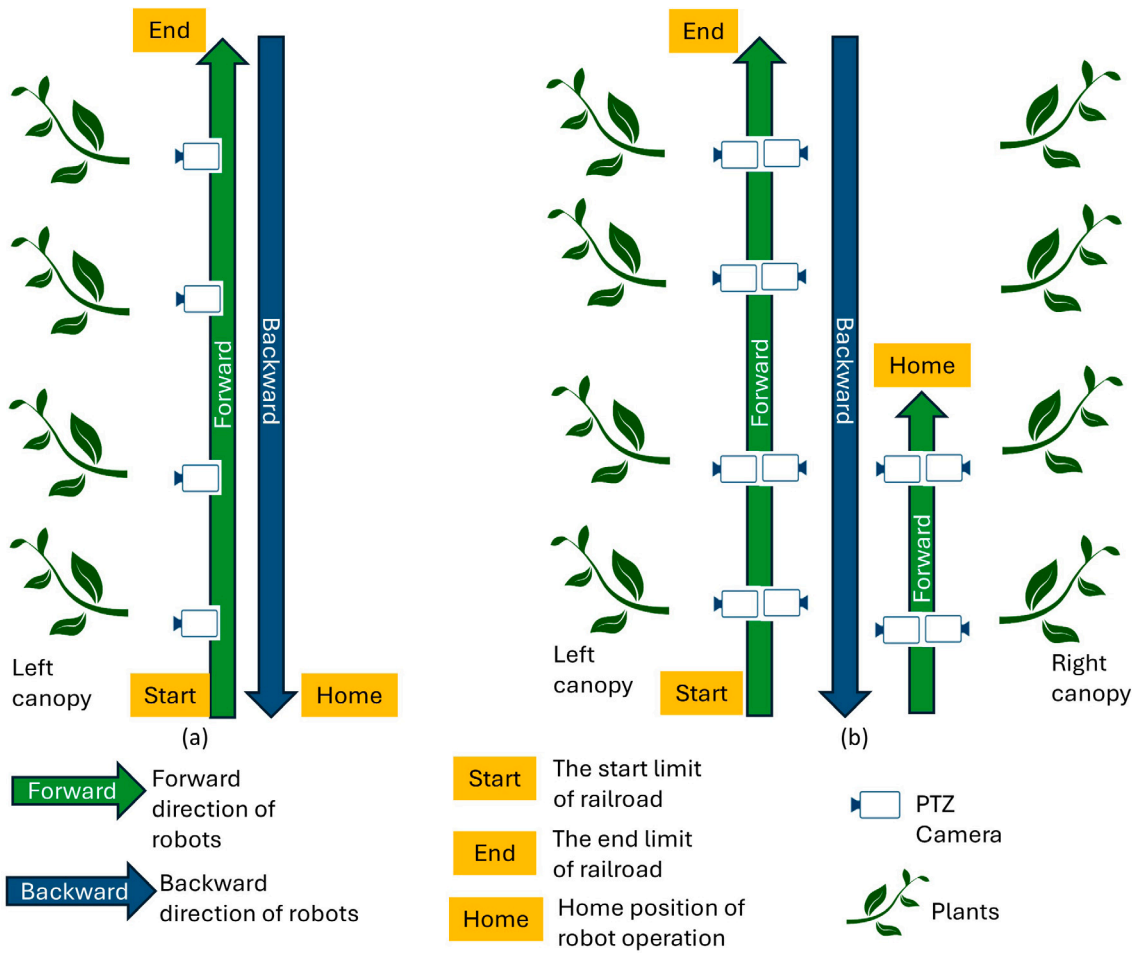


Fig. 6. Data acquisition procedure: (a) Ground-based robot, (b) Overhead robots.

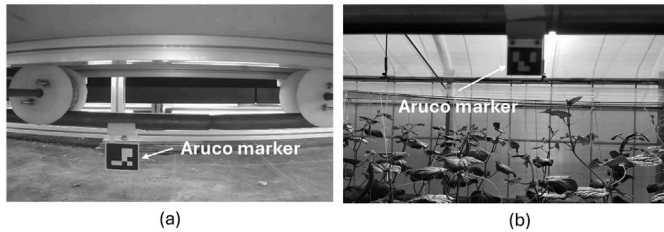


Fig. 7. Captured markers by IR camera under the railroad: (a) Ground-based robot, (b) Overhead robot.

As depicted in Table 1, there are two training datasets: training set #1, consisting of 700 original images (500 images collected by the ground-based robot, 200 images captured by the overhead robots), and training set #2, comprising 2150 images. The second dataset incorporates images from set #1, along with 1400 augmented data (rotated, flipped, brightness changes) and an additional 50 images with null annotations. Both training datasets utilized images captured within the zoom range of 1x to 10x. Two validation sets will be employed: validation set #1, composed of images featuring small plant details visualized at 0 to 2x zoom, and validation set #2, encompassing images showcasing extensive plant features visualized at a more excellent zoom range of 5 to 10x. Each validation set consists of 200 images. All labeled images were resized to 640×640 pixels. Additionally, Table 1 provides information on the number of images containing different detection categories. It is important to note that one image can contain multiple categories.

Table 1

Number of images in training sets #1 and #2, in validation sets #1 and #2, and images of different detection categories.

	Training set #1	Training set #2	Validation set #1	Validation set #2
Total images	700	2150	200	200
Original images	700	700	200	200
Augmented images	0	1400	0	0
Null annotation images	0	50	0	0
Bud images	176	528	64	45
Female flower images	249	747	82	66
Male flower images	174	522	58	39
Mature cucumber images	401	1203	95	59
Tiny cucumber images	273	819	76	52

Unit: number of images

2.2.3. Plant feature detection

In this investigation, the YOLOv8s model and the YOLOv7 normal model were selected for training in the detection of plant features. The models were trained via transfer learning (Zhuang et al., 2020) by fine-tuning pre-trained weights learned on the COCO dataset (Lin et al., 2014).

The plant feature detection models were trained using the PyTorch 2.1.0 deep learning framework on an Ubuntu 20.04 operating system. The training process involved utilizing CUDA version 11.8, with the Xeon W-2135 serving as the central processing unit (CPU) and the Titan-V was handling the graphics processing unit (GPU) tasks for image training.

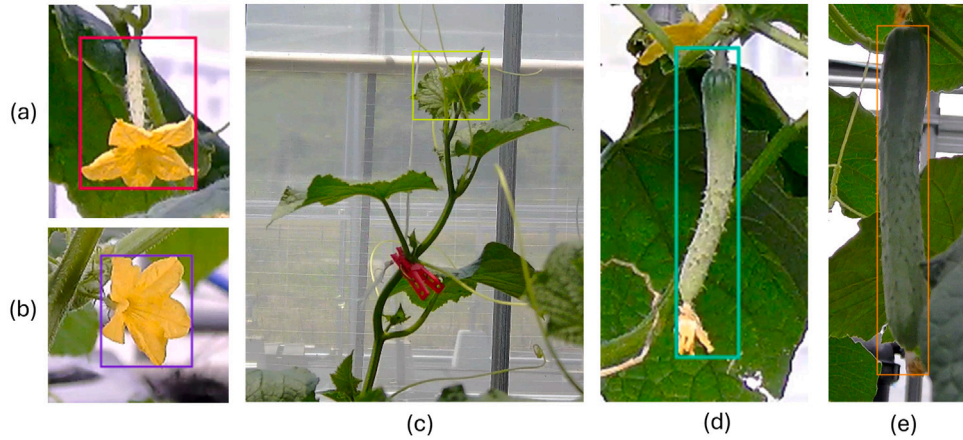


Fig. 8. Labeled classes: (a) Female flower, (b) Male flower, (c) Apical bud, (d) Tiny cucumber, (e) Mature cucumber. The bounding box zone in the images indicates the object's location.

2.2.4. Evaluation metrics

We employed Mean Average Precision (mAP) and the confusion matrix as the evaluation metrics to assess how well the trained models performed on the dataset.

The mAP, defined by Eq. (4), is determined by taking the mean of these AP values across all classes. AP in Eq. (3) is derived from precision and recall values obtained at various confidence thresholds for positive predictions. The precision–recall curve is generated by varying the confidence threshold, and at each threshold, the corresponding precision and recall values are calculated. The Area Under the Curve (AUC) of the precision–recall curve gives the Average Precision. Precision, defined by Eq. (1), is the ratio of true positives to the sum of true positives and false positives, while Recall, defined by Eq. (2), is the ratio of true positives to the sum of true positives and false negatives. True Positive (TP) denotes accurate detection of marked plant features, False Positive (FP) represents misidentified objects, and False Negative (FN) indicates overlooked plant features. Precision measures the proportion of correctly recognized plant features in the images in which the plant features were identified. At the same time, Recall (or recall rate) gauges the proportion of accurately identified plant features out of all images containing plant features. Statistical Metrics are calculated using the following equations (Wang et al., 2024):

$$\text{Precision} = \frac{TP}{TP + FP} \quad (1)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (2)$$

$$AP = \int_0^1 P(R) dR \quad (3)$$

$$mAP = \frac{\sum_{i=1}^N AP_i}{N} \quad (4)$$

$$\text{IoU} = \frac{\text{area}(gt \cap pd)}{\text{area}(gt \cup pd)} \quad (5)$$

The TP, FP, TN, and FN metrics are affected by the Intersection over the Union (IoU) threshold. IoU, also known as the Jaccard index, measures the similarity between the ground-truth mask (gt) and the predicted mask (pd). It is computed by dividing the area of overlap/intersection between gt and pd by the area of their union, as depicted in Eq. (5). This study selected an IoU threshold of 0.5, 0.6, and 0.7.

Fig. 9 displays the confusion matrix. It categorizes predictions into four quadrants: TP, FP, TN, and FN. This matrix provides a comprehensive evaluation, examining the performance of each class by considering both actual and predicted labels. Through a thorough matrix analysis, valuable insights are obtained regarding the model's

		Actual	
		Class 1 (Positive)	Class 1 (Negative)
Prediction	Class 1 (Positive)	TP	FP
	Class 1 (Negative)	FN	TN

Fig. 9. Confusion matrix.

efficacy on individual courses, allowing for identifying well-predicted classes and those requiring refinement. This level of detail facilitates targeted model optimization, focusing efforts on specific courses and ultimately improving the overall accuracy of the classification process.

2.3. Evaluation test

The evaluation process unfolded across several distinct steps. It commenced with the ArUco marker detection test, followed by examining the security switch's functionality. Subsequently, the performance of the reed switch mechanism was scrutinized, alongside an assessment of the PTZ camera's movement dynamics. Lastly, the evaluation encompassed the training of plant feature detection, culminating in a comprehensive evaluation process. The training approach includes evaluating the impact of incorporating zoomed images in model validation. Models are constructed using training sets #1 and #2, and subsequently, each trained model undergoes testing on validation sets #1 (focused on small feature visualization) and #2 (emphasizing big feature visualization).

3. Results

3.1. Fundamental operation of robots

The evaluation assessments were conducted at the Ito Plant Experiment Fields and Facilities, Faculty of Agriculture, Kyushu University, and the Kyushu University and Welzo Research Experiment Field, spanning from May 1st to July 15th, 2023. A group of five robots, comprising four overhead and one ground-based robot. The evaluation process involved steps such as ArUco marker detection, security switch functionality examination, scrutiny of the reed switch mechanism, assessment of the PTZ camera's movement dynamics.

Table 2 illustrates the success rates of AR marker detection across 100 paths. As depicted in Fig. 6, the ground-based robot follows a path

Table 2
Results of AR marker detection test on 100 paths.

	Ground-based robot	Overhead Robots			
	Robot #1	Robot #1	Robot #2	Robot #3	Robot #4
Number of tested paths	100	100	100	100	100
Number of AR markers/path	4	6	11	11	11
Number of tested AR marker detection times	400	600	1100	1100	1100
Number of successful AR marker detection	395	592	1098	1094	1095
Success rate (%)	98.75	98.67	99.82	99.45	99.55

involving forward movement from the starting point to the end of the railroad, then backward to the starting point (the home position is at the starting point). On the other hand, overhead robots move forward from home to the end of the railroad, return to the starting point, and finally proceed to the home position. The integrated IR cameras in the robots demonstrated a detection success rate of over 98% for ArUco markers positioned along the rail. This capability enables precise identification of both the home position and designated data collection points. Despite the high success rate, it is essential to acknowledge the occurrence of false alerts when the IR camera detects incorrect IDs, particularly in the intricate context of a complex plant canopy.

Furthermore, the mechanical switch underwent successful testing, effectively cutting off motor power in cases where the reed switches malfunctioned. This mechanism plays a pivotal role in ensuring system stability, preventing potential malfunctions that could lead to damage to either the plants or the robot.

To avoid the potential failure of reed switch recognition caused by the intensive processing of marker IDs, which significantly burdens the Raspberry Pi's processing capabilities and results in delays for other functions, particularly the reed switch recognition, magnetic field ranges have been increased at the start and end limits of the railroad. This implementation has worked successfully, ensuring the effective recognition of reed switches in the system.

Upon detecting the collection location, the automatic movement of the PTZ camera initiates, proceeding to preset viewpoints that have been set in advance. The specific angles and levels of zoom for each of these preset viewpoints vary according to user preferences. The PTZ camera system allows for up to 50 possible presets. Notably, throughout the study period, no reports were generated by the alert function indicating any issues with the execution of preset movements.

Throughout the study periods, the ground-based robot amassed a collection exceeding 10,000 images, whereas the overhead robots captured in excess of 65,000 images. An illustrative example of these collected images is depicted in Fig. 10.

The PTZ camera's pan, tilt and zoom feature is advantageous for capturing plant images as seen in Fig. 11. At 1x zoom (Fig. 11a), it offers an overview of plant scenes. Next, the PTZ camera would pan and tilt to viewpoint #1, #2 and #3 in Fig. 11a and change to several zoom levels, 5x (Fig. 11b) and 10x (Fig. 11c and 11d), allowing for a detailed examination of mature fruit, tiny fruit and leaf.

In scenarios involving rapidly growing plants like cucumbers, capturing all growth features can be challenging due to their diverse growth directions. The application of pan-tilt functionality in image capture yielded superior results. This was exemplified by Fig. 12, which showed how images obtained through this mechanism offered a more comprehensive and textured view of plant features, extending beyond the limitations of conventional fixed cameras.

3.2. Plant feature detection analysis

3.2.1. Training effect

In the training process, the YOLOv8s model, with 11.2 million parameters, and the YOLOv7 normal model, with 36.9 million parameters, were both configured with an initial learning rate of 0.01. The training batch size was set to 8 images, and the total number of iterations was fixed at 150.

During the training phase on training set #1, the YOLOv7 normal model required 80 min, while the YOLOv8s model achieved training in 42 min. When transitioning to training set #2, the YOLOv7 normal model completed training in 5.7 h, while the YOLOv8s model demonstrated a faster training time, finishing in 2.8 h.

In terms of inference time (ms/image), the YOLOv7 normal model exhibited an inference time of 12 ms, whereas the YOLOv8s model achieved a slightly faster inference time of 7 ms.

The mean Average Precision (mAP) outcomes, obtained by utilizing training sets #1 (with augmentation and null annotation data) and #2 (without augmentation and null data), are presented in Table 3. The models were subsequently evaluated on validation sets #1 (depicting small visualizations of plant features) and #2 (featuring more extensive visualizations), employing IoU thresholds of 0.5, 0.6, and 0.7.

In general, the YOLOv8s model demonstrates superior mean Average Precision (mAP) results compared to the YOLOv7 model. This suggests that the YOLOv8s model is more effective in accurately detecting and localizing objects in the given dataset. Besides, there is a consistent decrease in the mean Average Precision (mAP) score as the Intersection over Union (IoU) threshold is raised from 0.5 to 0.6 and further to 0.7. The augmentation of the training sets with both augmented data and null annotation images, as indicated in Table 3, results in improvements in mean Average Precision (mAP) scores.

Comparing Table 3a and Table 3b, utilizing the same trained model, the mean Average Precision (mAP) score is more excellent when evaluated on validation set #2, where plant features appear larger in images, in contrast to the evaluation on validation set #1, where plant features appear smaller in images.

3.2.2. Analysis of tested results on different validation sets

Overall, the mean Average Precision (mAP) scores tested on the same trained model indicate better performance on validation set #2 compared to validation set #1.

As indicated in the final two columns of Table 3a and 3b, the evaluation of the YOLOv8s model, trained with training set #2 (T#2), consistently yields the highest mean Average Precision (mAP) scores across both the tables with IoU 0.7 and the following analysis in this subsection analyzing the detection results of this trained model on validation sets #1 and #2.

The comparison between tested results on validation set #1 and validation set #2 (with IoU 0.7) reveals notable improvements. Specifically, for the "bud" category, the mAP increases from 0.931 to 0.94.

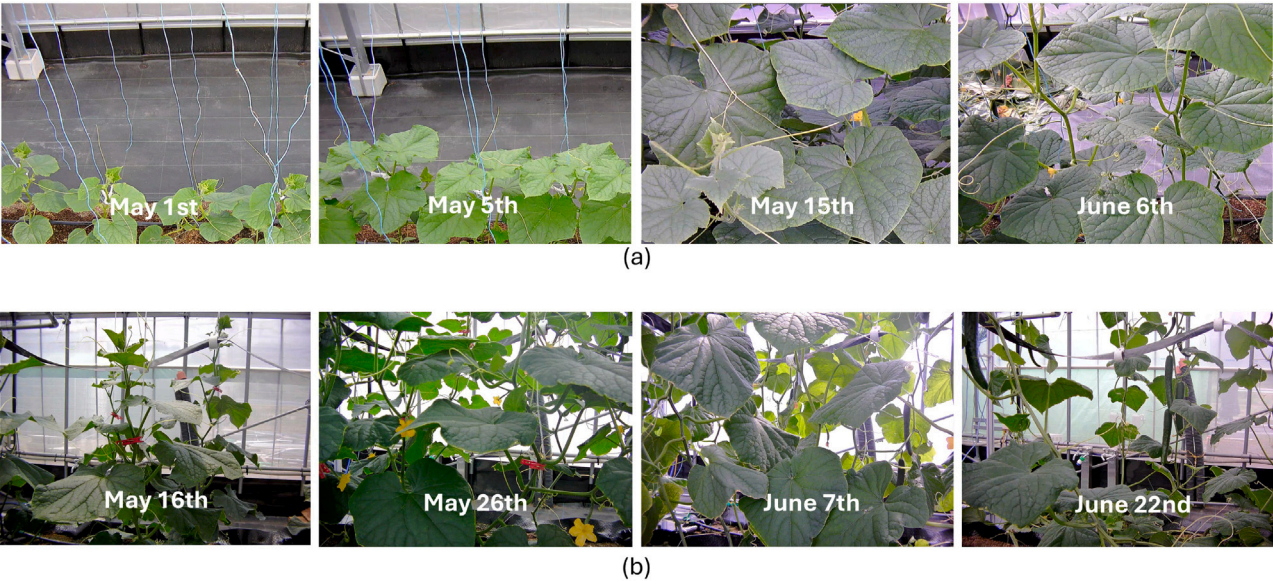


Fig. 10. Captured growth samples at a specific position: (a) By the overhead robots and (b) By the ground-based robot.

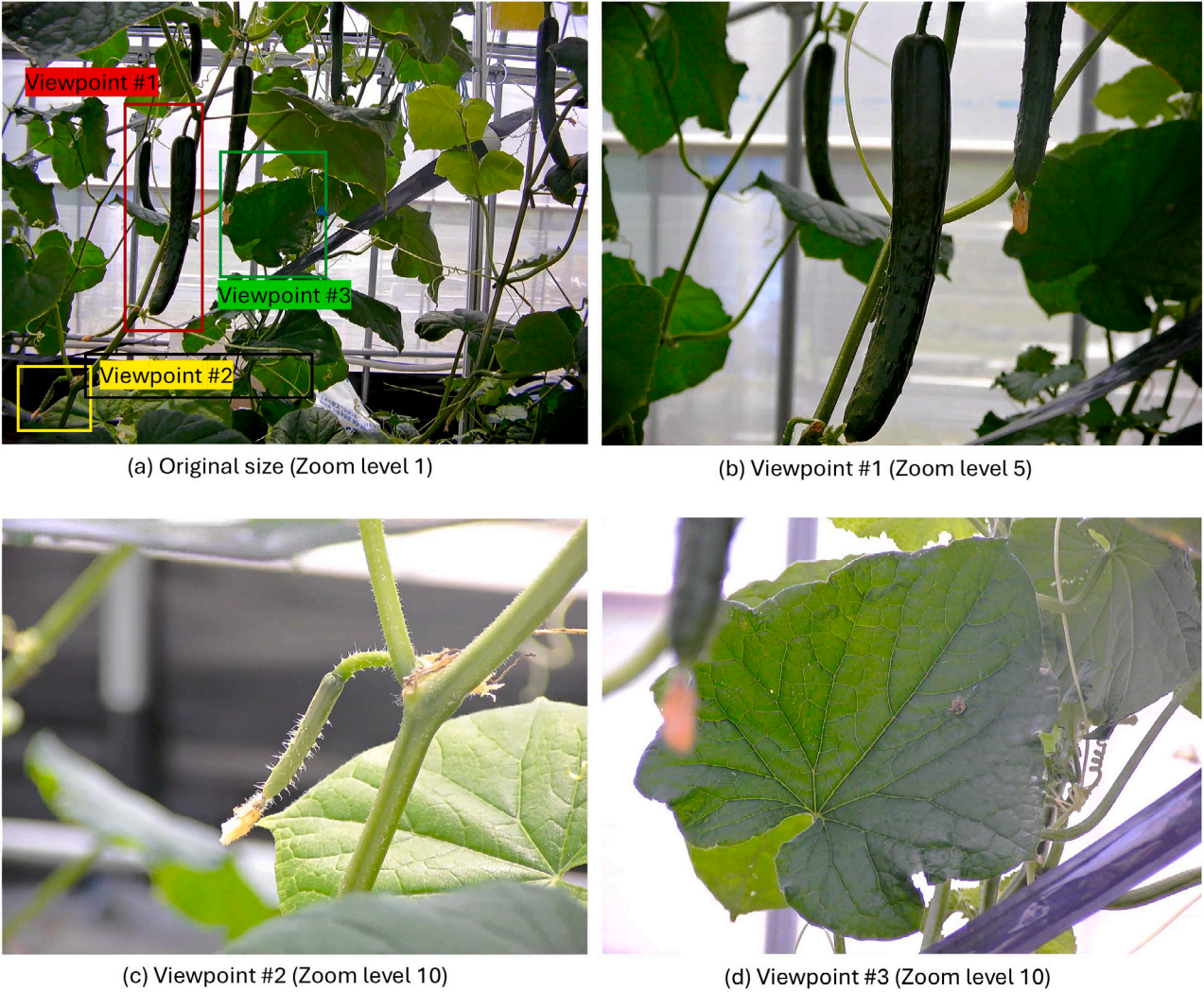


Fig. 11. Captured images using different zoom level of PTZ camera.



Fig. 12. Captured images using PTZ camera: (a) Growing point and (b) Plant root and soil.

Table 3
mAP Results of trained models on training set #1 and #2 for two validation sets: (a) Validation set #1, (b) Validation set #2.

	YOLOv7						YOLOv8s					
	IoU=0.5		IoU=0.6		IoU=0.7		IoU=0.5		IoU=0.6		IoU=0.7	
	T#1	T#2	T#1	T#2	T#1	T#2	T#1	T#2	T#1	T#2	T#1	T#2
Bud	0.901	0.93	0.89	0.905	0.887	0.89	0.915	0.939	0.913	0.938	0.91	0.931
Female flower	0.894	0.919	0.892	0.907	0.89	0.904	0.902	0.924	0.898	0.923	0.895	0.922
Male flower	0.794	0.824	0.788	0.821	0.785	0.818	0.793	0.834	0.793	0.83	0.791	0.827
Mature cucumber	0.834	0.819	0.828	0.812	0.815	0.806	0.811	0.804	0.808	0.792	0.805	0.791
Tiny cucumber	0.782	0.815	0.759	0.82	0.751	0.817	0.777	0.837	0.776	0.836	0.773	0.833

(a)

	YOLOv7						YOLOv8s					
	IoU=0.5		IoU=0.6		IoU=0.7		IoU=0.5		IoU=0.6		IoU=0.7	
	T#1	T#2	T#1	T#2	T#1	T#2	T#1	T#2	T#1	T#2	T#1	T#2
Bud	0.918	0.933	0.915	0.921	0.91	0.916	0.939	0.942	0.937	0.941	0.937	0.94
Female flower	0.946	0.96	0.94	0.957	0.936	0.94	0.953	0.987	0.953	0.985	0.95	0.984
Male flower	0.954	0.967	0.948	0.966	0.94	0.949	0.96	0.976	0.958	0.976	0.957	0.976
Mature cucumber	0.93	0.97	0.917	0.967	0.915	0.958	0.931	0.982	0.93	0.982	0.927	0.982
Tiny cucumber	0.814	0.898	0.81	0.894	0.8	0.881	0.85	0.909	0.847	0.909	0.83	0.901

(b)

T#1: training set #1 (without augmentation and null annotation data)
T#2: training set #2 (with augmentation and null annotation data)

The “female flower” category shows a substantial increase from 0.922 to 0.984, similar to the “male flower” category, which increases from 0.827 to 0.976. Additionally, there are significant improvements in the mAP for “mature cucumber”, rising from 0.791 to 0.982, and “tiny cucumber”, increasing from 0.833 to 0.901.

The confusion matrices for this trained model evaluated on validation set #1 and validation set #2 are shown in Figs. 13 and 14 respectively. As shown in Fig. 13, the “bud” and “female flower” classes demonstrate relatively good results with accuracies of 0.88 and 0.91, respectively. However, the remaining classes, namely “male flower”, “mature cucumber”, and “tiny cucumber”, exhibit lower accuracies at 0.72, 0.73, and 0.7, respectively. In Fig. 14, there are notable improvements across multiple classes. Specifically, the accuracies for “bud”, “female flower”, “male flower”, and “mature cucumber” increase substantially to 0.96, 0.96, 0.96, and 0.98, respectively. However, the accuracy for “tiny cucumber” sees a more modest improvement, reaching 0.76.

Fig. 15 displays instances of successfully detected plant features in validation set #1, whereas Fig. 16 shows examples where detection failed for male flowers, mature cucumbers, and tiny cucumbers.

Similarly, Fig. 17 illustrates successful detections of plant features in validation set #2, while Fig. 18 presents instances where the model failed to detect features of tiny cucumbers.

In summary, the results highlight the success of AR marker detection, surpassing a 98% accuracy rate and ensuring precise identification of specified points. Reed switch recognition and the implementation of mechanical switches for supporting robot navigation were also successful. The PTZ camera system, featuring versatile zoom and pan-tilt capabilities, autonomously captured a significant number of high-detail images, offering diverse perspectives on plant growth. The trained plant detection models demonstrated superior performance, especially when tested on validation sets with larger visualizations of plant features, evident in elevated mean average precision (mAP) scores and a well-performing confusion matrix.

4. Discussion

The success achieved in AR marker detection, reed switch recognition for identifying railroad limits, and the implementation of mechanical switches to prevent potential system damage to the plant

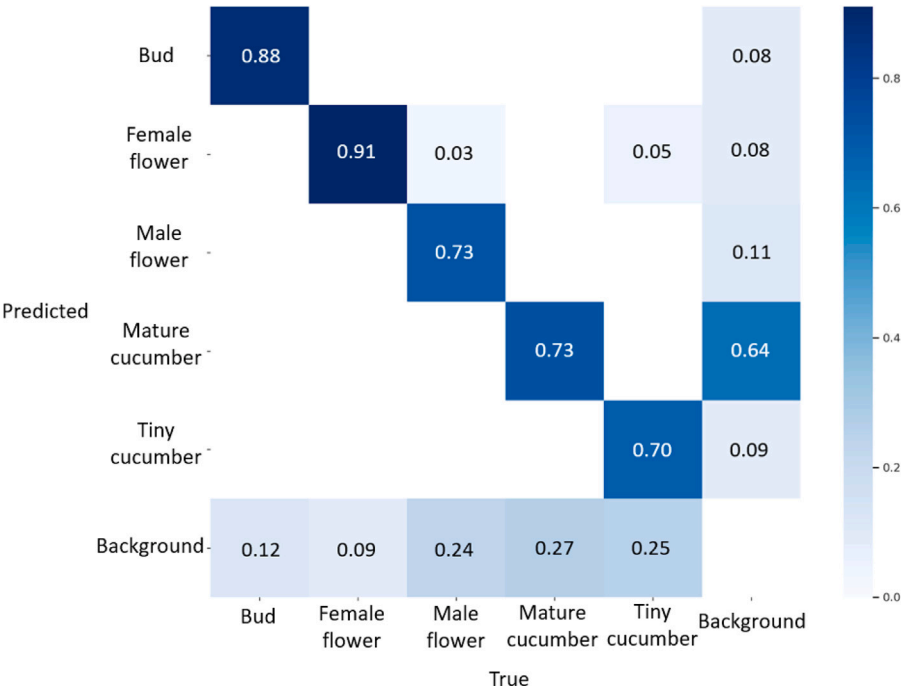


Fig. 13. Outcome confusion matrix when the trained YOLOv8s model using training set #2 is tested on validation set #1.

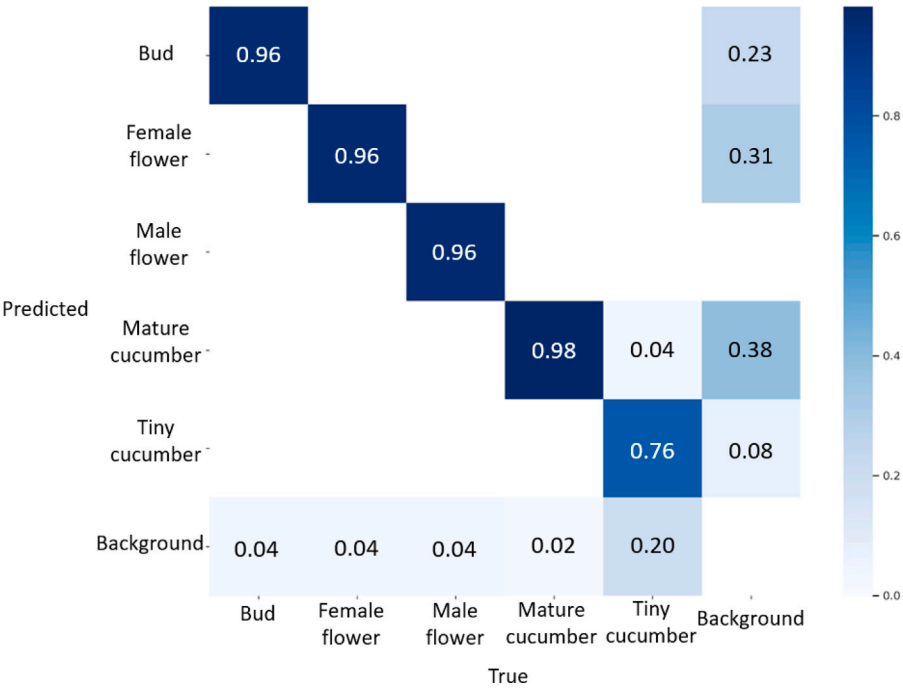


Fig. 14. Outcome confusion matrix when the trained YOLOv8s model using training set #2 is tested on validation set #2.

environment all demonstrated resilient navigation capabilities across critical aspects. These achievements signify a superior performance compared to the previous system, which relied on a stationary PTZ camera in precision agriculture (Kim et al., 2020). The PTZ camera exhibited excellent image capture capabilities. The application of pan-tilt functionality demonstrated a clear advantage over fixed cameras in capturing complex growth scenarios. The ability to adjust the camera's orientation and capture different angles offered

a more textured view of plant features. Besides, the integration of zoom capabilities within the PTZ camera system emerged as a crucial advancement. The detailed and vibrant images captured at various zoom levels provided valuable insights into small plant features. The ability to zoom in allowed for a closer examination of intricate growth features, which would have been challenging using standard cameras. These advantages underscore the impressive potential of PTZ cameras in collecting high-quality plant images, surpassing the capabilities of

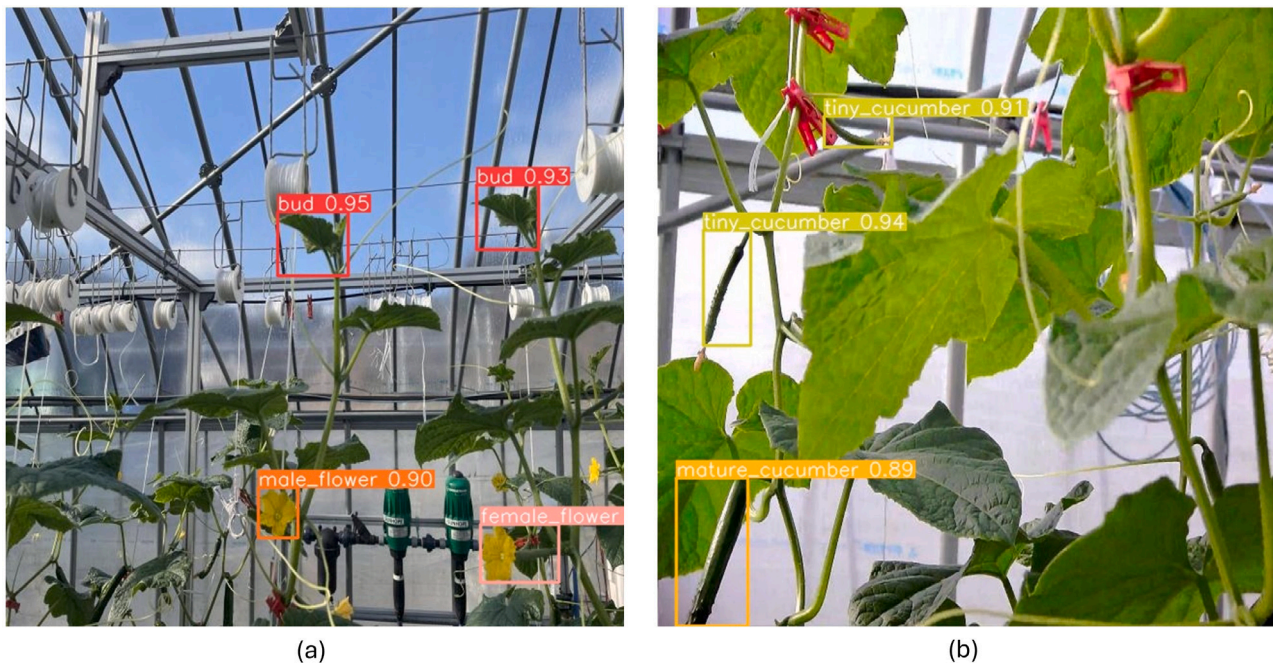


Fig. 15. Successful detected plant features in the validation set #1: (a) Detected bud, male and female flower, (b) Detected tiny and mature cucumber.

previous research in the field of plant phenotyping (Fukada et al., 2023). Notably, this study successfully implemented the automatic control of the PTZ camera using Raspberry Pi through ONVIF protocol, showing advancements over prior research that involved human intervention in controlling the PTZ camera plant phenotyping system.

In this study, the robot's effectiveness is enhanced despite following a straight path by integrating successful PTZ camera functions (pan, tilt, zoom, and preset viewpoint) to monitor the plant canopy. The expanded monitoring range provides advantages over fixed cameras, as the latter requires more installations in different locations and angles, resulting in complex camera management and potential obstacles in the plant-growing environment. The success of this approach establishes a robust foundation for future applications seeking a cost-effective plant phenotyping system.

By associating the captured images with precise time and location information, farmers gained the capacity to assess the status of plant growth in a comprehensive manner. The time-related data facilitated the tracking of growth progression at specific intervals, while the location data contributed insights into the prevailing environmental conditions and microclimates. This holistic dataset empowered farmers to monitor growth patterns, refine their management strategies, and proactively address potential risks. Consequently, these efforts culminated in heightened productivity within their agricultural pursuits.

The YOLOv8s model demonstrates shorter training and inference times compared to YOLOv7. Furthermore, as shown in Table 3, the mean Average Precision (mAP) results tested on YOLOv8s surpass those of YOLOv7 on our dataset. These findings strongly indicate a transition to YOLOv8s for our ongoing research efforts. Additionally, the enhanced mAP scores resulting from the utilization of augmentation and null annotation data underscore the positive impact of introducing diverse and informative examples in this research. It is noteworthy that the mAP score tends to increase as the Intersection over Union (IoU) decreases, suggesting that as the evaluation criteria become more stringent, demanding higher overlap between predicted and ground truth bounding boxes, the model's overall performance, measured by mAP, tends to decrease.

As indicated in Table 3, the higher mean Average Precision (mAP) score when testing the same model on validation set #2 compared

to validation set #1, along with the confusion matrix provided by Figs. 13 and 14, emphasizes the superior performance of the model on validation set #2. Fig. 16 emphasizes instances of failed plant feature detection on validation set #1, particularly for small visualizations (Fig. 16a and 16b) where male flowers appear diminutive and for indistinct features (Fig. 16c and d) when tiny and mature cucumbers are challenging to recognize due to their obscured positions within the complex plant canopy. In contrast, Fig. 17 highlights the successful detection of features on validation set #2, where the features exhibit more extensive visualizations in the images.

It is worth noting that the failed detection of tiny cucumbers in validation set #2 (Fig. 18) may be attributed to occlusion by other plant features, causing them to appear only partially in the images. Nevertheless, the overarching conclusion from these observations is that zoomed images positively impact the detection performance of the trained model on our dataset. This suggests that the model benefits from more detailed and expansive visual information, contributing to improved accuracy in scenarios with bigger visualizations.

Acknowledging the advantages of our system, it is important to note a significant limitation related to the inability to adjust the height of the camera. The recognition of the importance of adjusting camera height is particularly relevant for overhead robots that may encounter obstruction of the camera view by the rail, affecting the visibility of the root parts of the plant. However, it is crucial to emphasize that despite this limitation, the overhead robot can still capture the upper parts of the plant canopy. While the constraint exists, it does not diminish the core advantages of the system over previous research using PTZ camera in plant phenotyping (Kim et al., 2020; Fukada et al., 2023), especially in leveraging high-detailed images from the PTZ camera to enhance plant feature detection.

In summary, this research successfully implemented an automatic control system for utilizing a PTZ camera with pan-tilt-zoom functionality to capture diverse and highly detailed images of plants. The trained model demonstrated superior performance, particularly on images featuring zoomed-in plant features with high texture, showing advantages over previous research. Additionally, the study emphasizes the significance of Intersection over Union (IoU) on the detection results of the trained model.

Combining these findings, the research proposes further exploration, specifically examining the performance of a model trained with

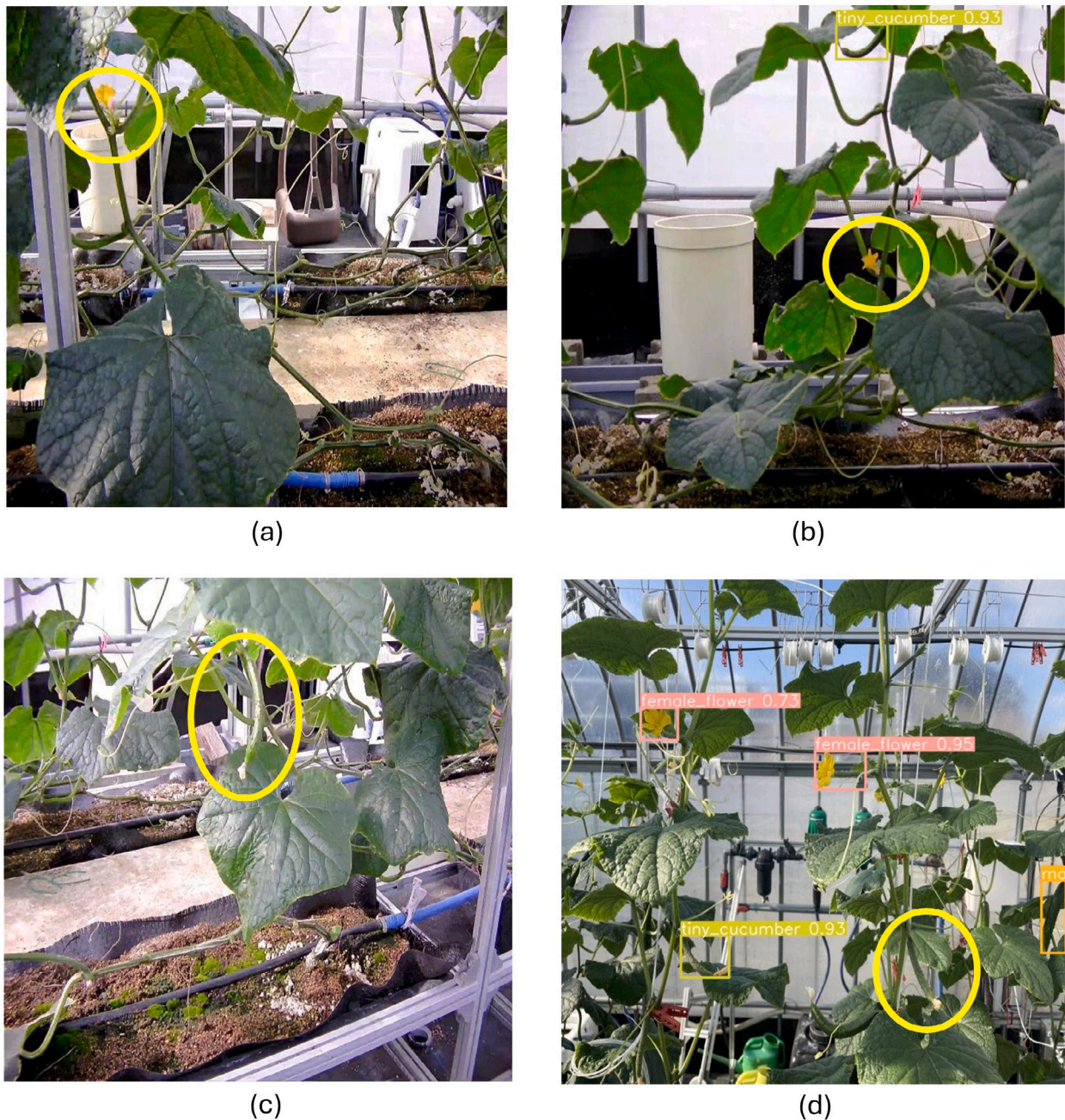


Fig. 16. Failed detected plant features in the validation set #1: (a) and (b) Male flower, (c) Mature cucumber, (d) Tiny cucumber (yellow oval shows failed detected plant feature location).

a low IoU on images with small visualizations of features and minimal zoom levels from the PTZ camera. Leveraging the automatic control of the PTZ camera's pan, tilt, and zoom functions could enhance feature detection, particularly when the plant feature is around the center of images and is zoomed in to obtain a larger visualization. Subsequently, applying a model trained with a high IoU ensures accurate and trustworthy plant feature detection.

5. Conclusion

In conclusion, this research successfully introduces a scalable high-throughput plant phenotyping technique utilizing a Pan Tilt Zoom (PTZ) camera, addressing the challenges faced by traditional automated phenotyping systems.

System reliability was validated through manual measurements and assessing the effectiveness of zoomed images captured by the PTZ camera. Experimental results showed promising outcomes, with a mean average precision (mAP) exceeding 90% for various plant traits in highly zoomed images, highlighting the efficacy of the proposed approach.

The findings emphasize the system's ability to capture real-time plant images with precision, showcasing the potential of the PTZ camera's zoom, pan, and tilt capabilities. The detailed visualization of plant traits contributes to improved results in plant feature detection.

In essence, this research not only introduces an innovative and effective plant phenotyping technique but also paves the way for advancements in automated phenotyping systems, offering scalability, precision, and adaptability to evolving plant growth scenarios. The

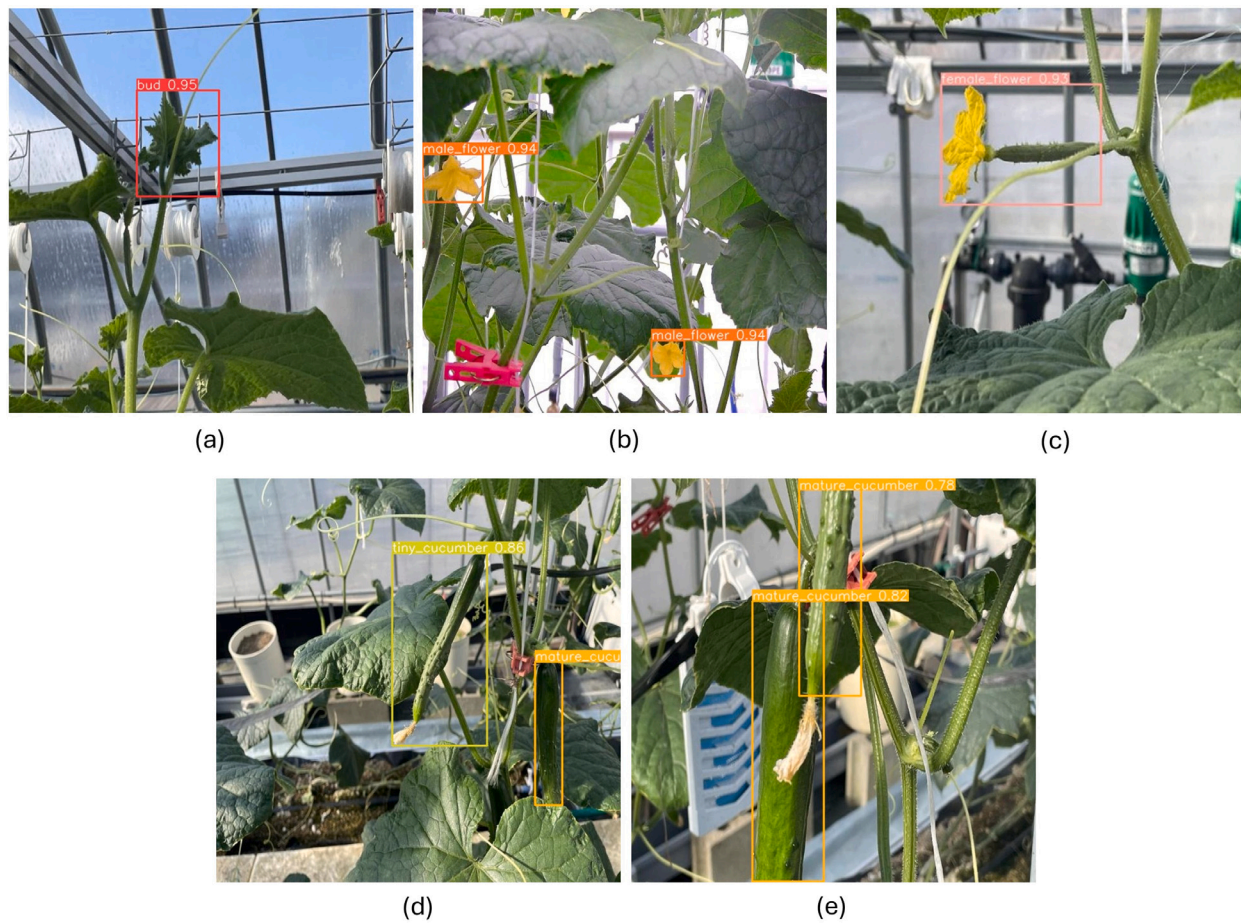


Fig. 17. Successfully detected plant features in the validation set #2: (a) Detected bud, (b) Detected male flower, (c) Detected female flower, (d) Detected tiny cucumber, (e) Detected mature cucumber.

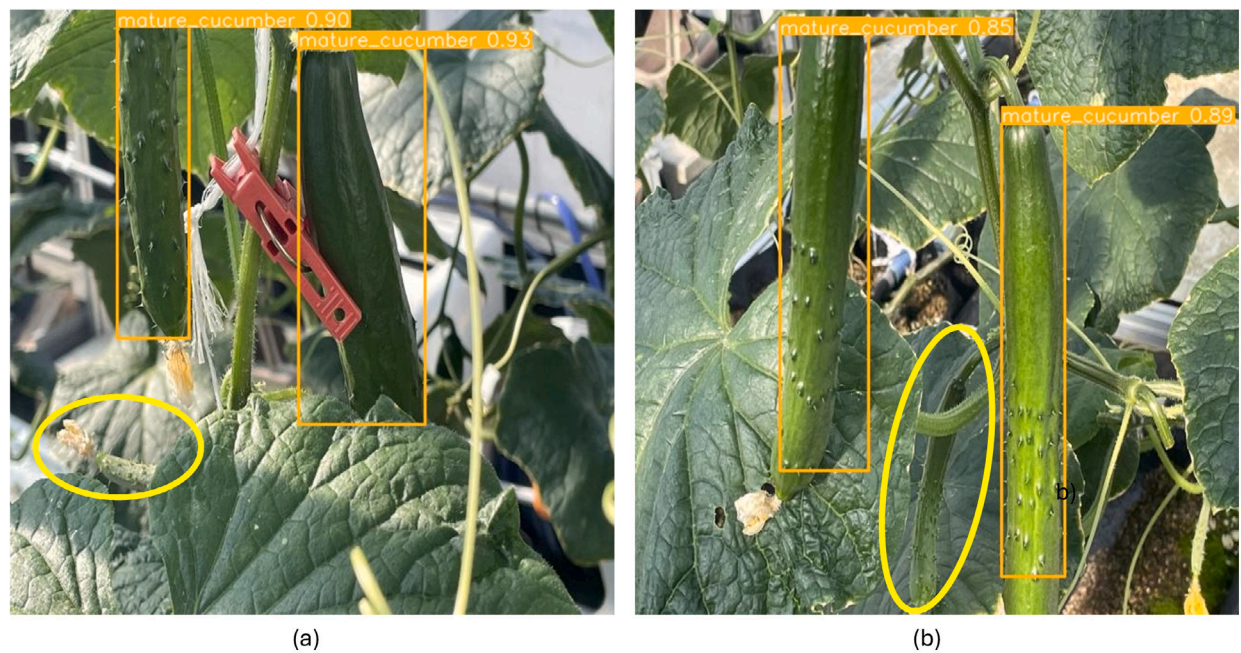


Fig. 18. Failed detected tiny cucumber in the validation set #2 (yellow oval shows failed detected plant feature location).

results affirm the significance of leveraging PTZ cameras in plant science and agriculture for enhanced quantitative analysis of plant growth attributes.

CRediT authorship contribution statement

Dong Thanh Pham: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Nayeen AI Amin:** Conceptualization, Methodology, Software, Validation. **Daisuke Yasutake:** Conceptualization, Methodology, Validation. **Yasumaru Hirai:** Conceptualization, Validation, Writing – review & editing. **Takenori Ozaki:** Funding acquisition, Investigation, Methodology. **Masaharu Koga:** Funding acquisition, Investigation, Methodology. **Kota Hidaka:** Funding acquisition, Investigation, Methodology. **Masaharu Kitano:** Conceptualization, Funding acquisition, Methodology, Validation. **Hien Bich Vo:** Methodology, Writing – review & editing. **Takashi Okayasu:** Supervision, Writing – original draft, Writing – review & editing, Conceptualization, Funding acquisition, Methodology, Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research was conducted as the collaborative research with the welzo Co. Ltd., the JSPS, KAKENHI No. JP21K05860, and the Cabinet Office grant in aid, the Advanced Next-Generation Greenhouse Horticulture by IoP (Internet of Plants), Japan. We want to express many thanks for their valuable support.

Data availability

The data that has been used is confidential.

References

- Amitrano, C., Junker, A., D'Agostino, N., De Pascale, S., De Micco, V., 2022. Integration of high-throughput phenotyping with anatomical traits of leaves to help understanding lettuce acclimation to a changing environment. *Planta* 256 (4), 1–19.
- Berger, B., de Regt, B., Tester, M., 2012. High-throughput phenotyping of plant shoots. In: *High-Throughput Phenotyping in Plants*. Springer, pp. 9–20.
- Bochkovski, A., Wang, C.-Y., Liao, H.-Y.M., 2020. YOLOv4: Optimal speed and accuracy of object detection. *arXiv preprint arXiv:2004.10934*.
- Bodner, G., Nakhforoosh, A., Arnold, T., Leitner, D., 2018. Hyperspectral imaging: a novel approach for plant root phenotyping. *Plant Methods* 14 (1), 1–17.
- Casto, A.L., Schuhl, H., Tovar, J.C., Wang, Q., Bart, R.S., Fahlgren, N., Gehan, M.A., 2021. Picturing the future of food. *Plant Phenome J.* 4 (1), e20014.
- Cramer, G.R., Urano, K., Delrot, S., Pezzotti, M., Shinozaki, K., 2011. Effects of abiotic stress on plants: a systems biology perspective. *BMC Plant Biol.* 11 (1), 1–14.
- FAO, 2018. The future of food and agriculture—alternative pathways to 2050. Rome: FAO. Licence: CC BY-NC-SA 3.0 IGO.
- Fukada, K., Hara, K., Cai, J., Teruya, D., Shimizu, I., Kuriyama, T., Koga, K., Sakamoto, K., Nakamura, Y., Nakajo, H., 2023. An automatic tomato growth analysis system using YOLO transfer learning. *Appl. Sci.* 13 (12), 6880.
- Ge, Y., Bai, G., Stoerger, V., Schnable, J.C., 2016. Temporal dynamics of maize plant growth, water use, and leaf water content using automated high throughput RGB and hyperspectral imaging. *Comput. Electron. Agric.* 127, 625–632.
- Han, W., Cui, L., Chen, W., Li, M., Su, K., Wu, P., 2014. Design of movable remote crop monitoring system on fixed truss. *Trans. Chin. Soc. Agric. Eng.* 30 (13), 160–168.
- Harakannanavar, S.S., Rudagi, J.M., Puranikmath, V.I., Siddiqua, A., Pramodhini, R., 2022. Plant leaf disease detection using computer vision and machine learning algorithms. *Global Transit. Proc.* 3 (1), 305–310.
- Hilty, J., Muller, B., Pantin, F., Leuzinger, S., 2021. Plant growth: the what, the how, and the why. *New Phytol.* 232 (1), 25–41.
- Jocher, G., Nishimura, K., Mineeva, T., Vilariño, R., 2020. YOLOv5 (2020). GitHub repository: <https://github.com/ultralytics/yolov5>.
- Johnson, N.C., Chaudhary, V.B., Hoeksema, J.D., Moore, J.C., Pringle, A., Umbanhowar, J.A., Wilson, G.W., 2009. Mysterious mycorrhizae? A field trip & classroom experiment to demystify the symbioses formed between plants & fungi. *Am. Biol. Teacher* 71 (7), 424–429.
- Kefauver, S.C., El-Haddad, G., Vergara-Diaz, O., Araus, J.L., 2015. RGB picture vegetation indexes for high-throughput phenotyping platforms (HTPPs). In: *Remote Sensing for Agriculture, Ecosystems, and Hydrology XVII*, Vol. 9637. SPIE, pp. 82–90.
- Kim, W.-S., Lee, D.-H., Kim, Y.-J., 2020. Machine vision-based automatic disease symptom detection of onion downy mildew. *Comput. Electron. Agric.* 168, 105099.
- Kumar, M., Pal, Y., Gangadharan, S.M.P., Chakraborty, K., Yadav, C.S., Kumar, H., Tiwari, B., 2022. Apple sweetness measurement and fruit disease prediction using image processing techniques based on human-computer interaction for industry 4.0. *Wirel. Commun. Mob. Comput.* 2022.
- Li, S., Fu, X., Dong, J., 2022. Improved ship detection algorithm based on YOLOX for SAR outline enhancement image. *Remote Sens.* 14 (16), 4070.
- Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., Zitnick, C.L., 2014. Microsoft coco: Common objects in context. In: *Computer Vision—ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6–12, 2014, Proceedings, Part V 13*. Springer, pp. 740–755.
- Mann, H.M., Ioannidis, A., Jepsen, J.U., Welker, J.M., Loonen, M.J., Høye, T.T., 2022. Automatic flower detection and phenology monitoring using time-lapse cameras and deep learning. *Remote Sens. Ecol. Conservat.* 8 (6), 765–777.
- Mertens, S., Verbracken, L., Sprenger, H., Demuyne, K., Maleux, K., Cannoot, B., De Block, J., Maere, S., Nelissen, H., Bonaventure, G., et al., 2021. Proximal hyperspectral imaging detects diurnal and drought-induced changes in maize physiology. *Front. Plant Sci.* 12, 640914.
- Mochida, K., Koda, S., Inoue, K., Hirayama, T., Tanaka, S., Nishii, R., Melgani, F., 2019. Computer vision-based phenotyping for improvement of plant productivity: a machine learning perspective. *GigaScience* 8 (1), giy153.
- Muramatsu, Y., Kubota, S., 2021. The development of a root-zone environmental control system (n. RECS) and its application to flower production. *Horticulture J.* 90 (3), 239–246.
- Peng, Y., Wang, Y., 2022. Leaf disease image retrieval with object detection and deep metric learning. *Front. Plant Sci.* 13.
- Prasanna, B.M., Araus, J.L., Crossa, J., Cairns, J.E., Palacios, N., Das, B., Magorokosho, C., 2013. High-throughput and precision phenotyping for cereal breeding programs. In: *Cereal Genomics II*. Springer, pp. 341–374.
- Ray, D.K., Mueller, N.D., West, P.C., Foley, J.A., 2013. Yield trends are insufficient to double global crop production by 2050. *PLoS One* 8 (6), e66428.
- Redmon, J., Divvala, S., Girshick, R., Farhadi, A., 2016. You only look once: Unified, real-time object detection. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. pp. 779–788.
- Redmon, J., Farhadi, A., 2017. YOLO9000: better, faster, stronger. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. pp. 7263–7271.
- Redmon, J., Farhadi, A., 2018. YOLOv3: An incremental improvement. *arXiv preprint arXiv:1804.02767*.
- Rouphael, Y., Kyriacou, M.C., Petropoulos, S.A., De Pascale, S., Colla, G., 2018. Improving vegetable quality in controlled environments. *Sci. Horticul.* 234, 275–289.
- Shi, Z., Guo, B., Zhao, M., Zhang, C., et al., 2018. Nighttime low illumination image enhancement with single image using bright/dark channel prior. *EURASIP J. Image Video Process.* 2018 (1), 1–15.
- Singh, A., Jones, S., Ganapathysubramanian, B., Sarkar, S., Mueller, D., Sandhu, K., Nagasubramanian, K., 2021. Challenges and opportunities in machine-augmented plant stress phenotyping. *Trends Plant Sci.* 26 (1), 53–69.
- Su, D., Qiao, Y., Jiang, Y., Valente, J., Zhang, Z., He, D., 2023. AI, sensors and robotics in plant phenotyping and precision agriculture, volume II. *Front. Plant Sci.* 14, 1215899.
- Tanabata, T., Kodama, K., Hashiguchi, T., Inomata, D., Tanaka, H., Isobe, S., 2022. Development of a plant conveyance system using an AGV and a self-designed plant-handling device: A case study of DIY plant phenotyping. *Breeding Sci.* 72 (1), 85–95.
- Tian, Y., Yang, G., Wang, Z., Li, E., Liang, Z., 2020. Instance segmentation of apple flowers using the improved mask R-CNN model. *Biosyst. Eng.* 193, 264–278.
- Van Eeuwijk, F.A., Bustos-Korts, D., Millet, E.J., Boer, M.P., Kruijer, W., Thompson, A., Malosetti, M., Iwata, H., Quiroz, R., Kuppe, C., et al., 2019. Modelling strategies for assessing and increasing the effectiveness of new phenotyping techniques in plant breeding. *Plant Sci.* 282, 23–39.
- Wang, C.-Y., Bochkovski, A., Liao, H.-Y.M., 2022. YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors. *arXiv preprint arXiv:2207.02696*.
- Wang, Y., Chu, M., Kang, X., Liu, G., 2024. A deep learning approach combining DeepLabV3+ and improved YOLOv5 to detect dairy cow mastitis. *Comput. Electron. Agric.* 216, 108507.
- Will, R.E., Wilson, S.M., Zou, C.B., Hennessey, T.C., 2013. Increased vapor pressure deficit due to higher temperature leads to greater transpiration and faster mortality during drought for tree seedlings common to the forest–grassland ecotone. *New Phytol.* 200 (2), 366–374.

- Xiao, B., Nguyen, M., Yan, W.Q., 2023. Fruit ripeness identification using YOLOv8 model. *Multimedia Tools Appl.* 1–18.
- Xiong, C., Hu, S., Fang, Z., 2022. Application of improved YOLOV5 in plate defect detection. *Int. J. Adv. Manuf. Technol.* 1–13.
- Yang, W., Feng, H., Zhang, X., Zhang, J., Doonan, J.H., Batchelor, W.D., Xiong, L., Yan, J., 2020. Crop phenomics and high-throughput phenotyping: past decades, current challenges, and future perspectives. *Molecular Plant* 13 (2), 187–214.
- Yang, T., Zhou, S., Xu, A., Ye, J., Yin, J., 2023. An approach for plant leaf image segmentation based on YOLOV8 and the improved DEEPLABV3+. *Plants* 12 (19), 3438.
- Zhao, C., Zhang, Y., Du, J., Guo, X., Wen, W., Gu, S., Wang, J., Fan, J., 2019. Crop phenomics: current status and perspectives. *Front. Plant Sci.* 10, 714.
- Zhuang, F., Qi, Z., Duan, K., Xi, D., Zhu, Y., Zhu, H., Xiong, H., He, Q., 2020. A comprehensive survey on transfer learning. *Proc. IEEE* 109 (1), 43–76.